

FFmpeg: high quality animated GIF?

Asked 7 years, 5 months ago Modified 4 years, 3 months ago Viewed 62k times

I'm generating animated a GIF from a video on my server.
The generated GIF is not really high quality and it looks like the pixels are huge.

Example:



This is how I generate the GIF:

```
shell_exec("/usr/bin/ffmpeg -i video.mkv -vf scale=500:-1 -t 10 -r 10 image.gif");
```

I did a search on Google and came across this:

```
shell_exec("/usr/bin/ffmpeg -i video.mkv -r 20 -f image2pipe -vcodec ppm - | convert -delay 5 - output.gif");
```

But the command above doesn't do anything and no *output.gif* is being generated at all.

There are some tutorials that I came across but none of them worked for me and some of them involve using ImageMagick which I dont have access to.

Could someone please let me know if there is a clear way to generate a high-quality GIF using FFmpeg?

animation ffmpeg Edit tags

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edited Jun 9, 2020 at 9:12

Matthias Braun
33.7k 26 151 176

asked Mar 23, 2017 at 15:37

David Hope
1,526 4 24 53

- [blog.pkh.me/p/21-high-quality-gif-with-ffmpeg.html](#) You can try playing with the dither and palettegen parameters here. – Aleksandar Misich Mar 23, 2017 at 15:43
- @AleksandarMisich, that is the first thing i came across in google.. however, as i stated in my question, i do not have access to imagemagic so that tutorial is not good for me.. also, the usage doesn't really make sense to me.. – David Hope Mar 23, 2017 at 15:45
- You don't need imagemagick for that tutorial. – Gyan Mar 23, 2017 at 16:30
- @Mulvy, correct. my bad... but i still don't understand the usage. especially because I'm using command lines to run ffmpeg. – David Hope Mar 23, 2017 at 16:56
- 1 See [How do I convert a video to GIF using ffmpeg, with reasonable quality?](#) – llogan Mar 23, 2017 at 18:57

4 Answers

Sorted by:

Date modified (newest first)

Reset to default

`shell_exec("/usr/bin/ffmpeg -i video.mkv -r 20 -f image2pipe -vcodec ppm - | convert -delay 5 - output.gif");`

I suppose you have no [imageMagick](#) installed on your environment, because "convert" is one of IM's tools.

As for the video artifacts, it is caused by the default dithering method in FFmpeg. For best results, I'd recommend `floyd_steinberg` or `sierra2_4a`, and maybe `bayer` with scale set to 3. (Also, there's no such things like "huge" pixels, they are the atomic elements of raster images.)

On the other side, you can achieve better results with ffmpeg only. First, I'd generate a palette of the input video:

```
ffmpeg -i <your_input.mkv> -filter_complex "fps=10;scale=500:-1:flags=lanczos,palettegen=stats_mode=full" -t 10 palette.png
```

Then, use this color template to generate the actual gif file:

```
ffmpeg -i <your_input.mkv> -i palette.png -filter_complex "[0]fps=10;scale=500:-1:flags=lanczos[scaled]; [scaled][1:v] paletteuse=dither=sierra2_4a" -t 10 <output.gif>
```

You might need to fiddle with the params and the dithering methods to achieve best result. You may also try to generate new palette for each frame, so you can skip the first pass, and use the `new` option in the `paletteuse` filter.

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edited Nov 28, 2019 at 14:57

answered Mar 24, 2017 at 9:48

Gergely Lukacsy
3,034 2 25 29

- 7 Just a note: you can combine the two commands into one using a slightly more complex filtergraph: `ffmpeg -i <your_input.mkv> -filter_complex "fps=10;scale=500:-1:flags=lanczos,split[v1][v2]; [v1]palettegen=stats_mode=full [palette];[v2]palette]paletteuse=dither=sierra2_4a" -t 10 <output.gif>` – paradox460 Sep 12, 2019 at 19:15

- @paradox460 I think your comment should be an answer to make it more visible – [Martin Castin](#) Oct 5, 2021 at 13:15
- 11 There apparently is a `[ missing before palette ]`. Reworking this command without `scale` and `t` plus introducing `-vsync 0` led to reasonably sized `gif` s here: `ffmpeg -y -i input.mkv -filter_complex "fps=15,split[v1][v2]; [v1]palettegen=stats_mode=full [palette]; [v2][palette]paletteuse=dither=sierra2_4a" -vsync 0 output.gif` – [almereyda](#) Jul 8, 2022 at 17:06
- @almereyda your suggestion works perfect! – [kylron](#) Jan 20, 2023 at 22:35



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I've written a tool specifically for maximum quality:

<https://gif.ski>

```
ffmpeg -i video.mp4 frame%04d.png
gifski -o clip.gif frame*.png
```

It generates good per-frame palettes, but also combines palettes across frames, achieving even thousands of colors per frame.



If you want to reduce the video dimensions, add a scaling filter:

```
ffmpeg -i video.mp4 -vf scale=400:240 frame%04d.png
```

If you want to reduce the frame rate, add the `fps` filter:

```
ffmpeg -i video.mp4 -vf fps=12 frame%04d.png
```

You can combine the filters with `-vf scale=400:240,fps=12`

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edited Jan 10, 2019 at 16:33



[slhck](#)

38.1k

32

157

212

answered Nov 25, 2017 at 13:34



[Kornel](#)

99.6k

38

231

316

2 Awesome work. I'm exploring imageoptim as a result of seeing the quality of gifski. – [Mahmoud Al-Qudsi](#) Sep 5, 2018 at 18:15



That color space is beautiful, but the resulting file size in my quick test is 3-4x what I get with ffmpeg's default gif encoder. Am I doing something wrong? I'm using the latest Windows exe available from GitHub releases (version 0.8.5). – [TigerhawkT3](#) Oct 25, 2019 at 4:16



It's normal that higher-quality file takes much more space. The best way to reduce gif size is to reduce motion (area that changes between frames) or framerate. You can also try giflossy (`gifsicle --lossy`). – [Kornel](#) Oct 26, 2019 at 20:08

1 doesn't look like CLI works for the windows exe, it just opens the GUI – [Hashbrown](#) Mar 18 at 8:30



1 Fast and nice result. Superior to the `convert` and `FFmpeg palette` methods. – [rdupz](#) Apr 8 at 21:03

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The key issue is that any gif picture or frame has an **extremely limited palette** of only 256 of the possible millions of colors in your video.

[This is well explained here.](#)

So, *fairly recently* (2015, version 2.6) ffmpeg got the **palettegen** and **paletteuse** filters that can **generate better palettes for each frame**.

Therefore, make sure you are using a *fairly recent* version of ffmpeg.

So, there's your secret and key search term to get you to make high quality gifs in no time - study up on **palettegen** filters. [Reddit beware.](#)

Some references:

[ffmpeg 2.6 release notes](#)

[ffmpeg docs](#)

[superuser](#)

[blog.phk.me](#)

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edited May 5, 2017 at 20:10

answered Mar 24, 2017 at 1:02



[neonzeon](#)

207

1

5



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This post is hidden. It was [deleted](#) 7 years ago by the post author.

```
ffmpeg -i fileIn.flv -vf scale=320:-1 -r 10 -f image2pipe -vcodec ppm - | convert -delay 5 -loop 0 - fileOut.gif
```



Convert video to gif by ffmpeg and imagemagick you can use any common video format. if you don't need to change the size of gif output, just remove `-vf scale=320:-1` btw, 320:-1 means width is 320px and height would be set automatically

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answered May 23, 2017 at 14:57



Gerald Hughes

6,099 20 77 139

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